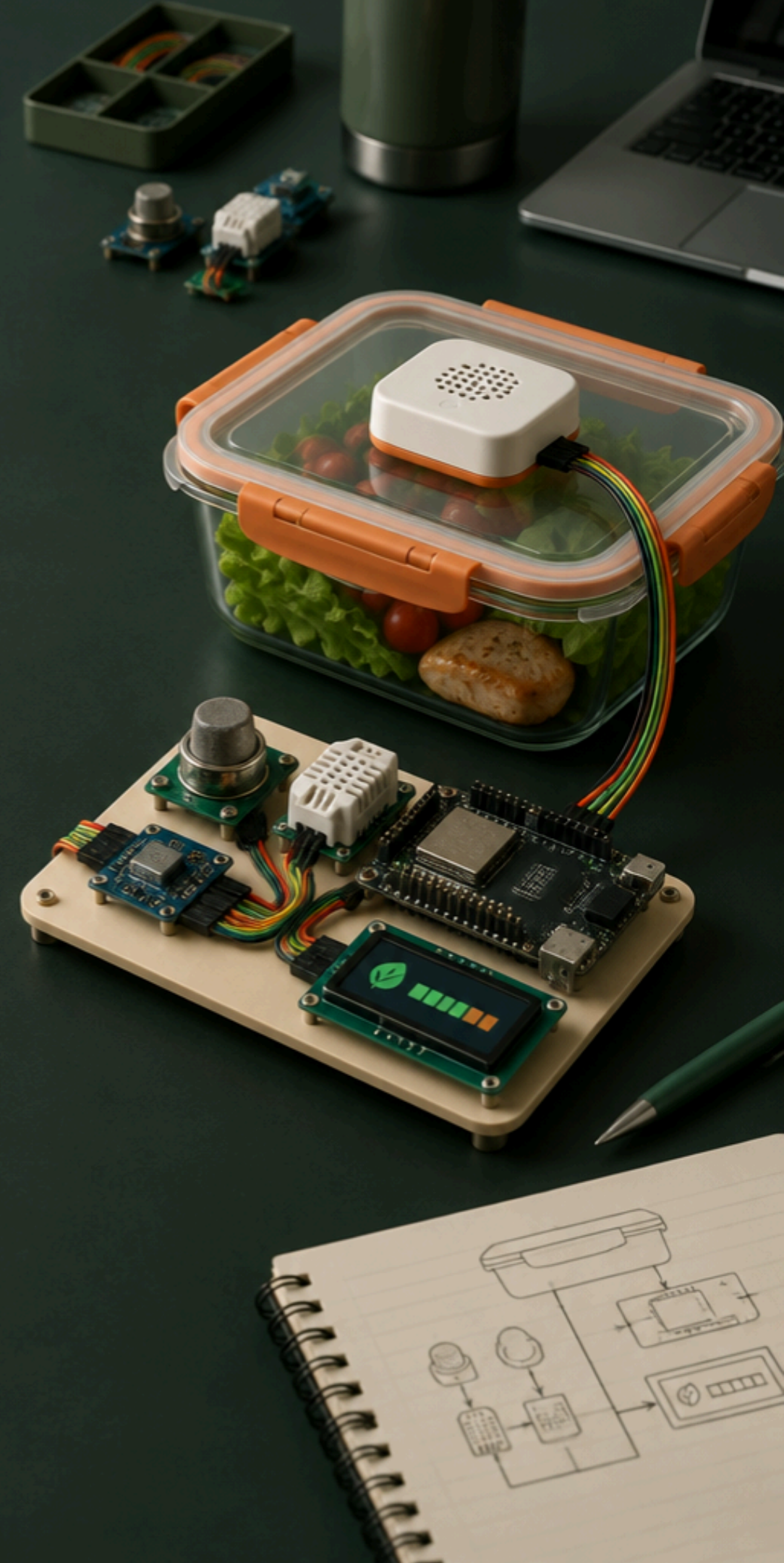


DEVELOPER GUIDE • SRS-ALIGNED

FreshGuard

IoT-Based Smart Food Freshness Monitor

Professional academic implementation guide
Team Inferno-Net



Project Overview

Purpose, boundaries and success criteria

PURPOSE

Monitor food condition using environmental and gas-sensing inputs.

OBJECTIVES

Acquire sensor data; identify food; estimate freshness; notify users.

IN SCOPE

Prototype sensing, local status, dashboard and temporary offline operation.

LIMITATIONS

Not a laboratory-grade safety instrument. Exact accuracy depends on calibration.

System Architecture

End-to-end implementation architecture



Hardware Components

Inventory derived from SRS roles

COMPONENT / ROLE	SELECTION STATUS
Controller — acquisition, logic, connectivity	To be selected by team
Gas sensor(s) — spoilage-related signal	Model to be selected by team
Temperature / humidity — context input	Model to be selected by team
Display / LEDs / buzzer — local feedback	As specified or team-selected
Power, breadboard, cables, enclosure	Document ratings and quantities

Software Requirements

Development environment and implementation options

FIRMWARE

IDE, language and libraries:
select to match controller.

CONNECTIVITY

Protocol and network
credentials: to be selected by
team.

DATA

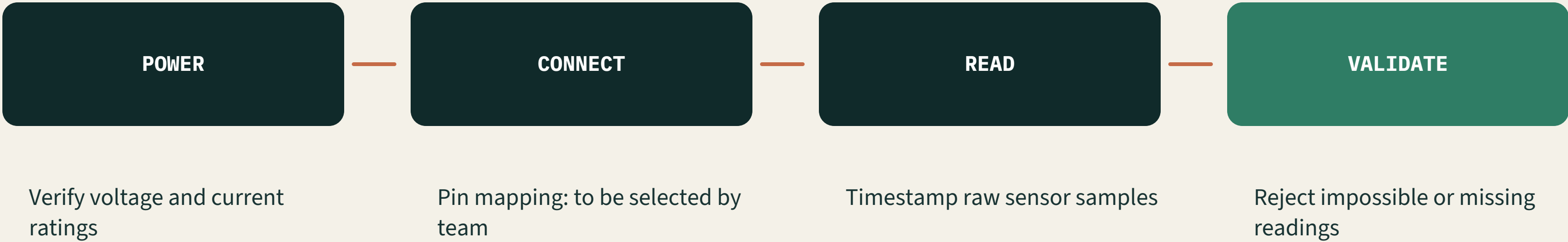
Local buffer plus remote
persistence; define schema and
timestamps.

INTERFACE

Dashboard technology: use an
SRS-listed option; document
versions.

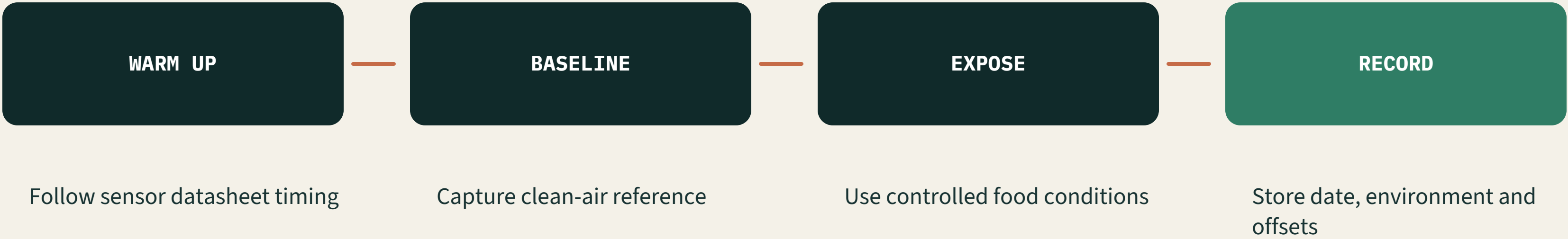
Hardware & Sensor Integration

Safe wiring and traceable data flow



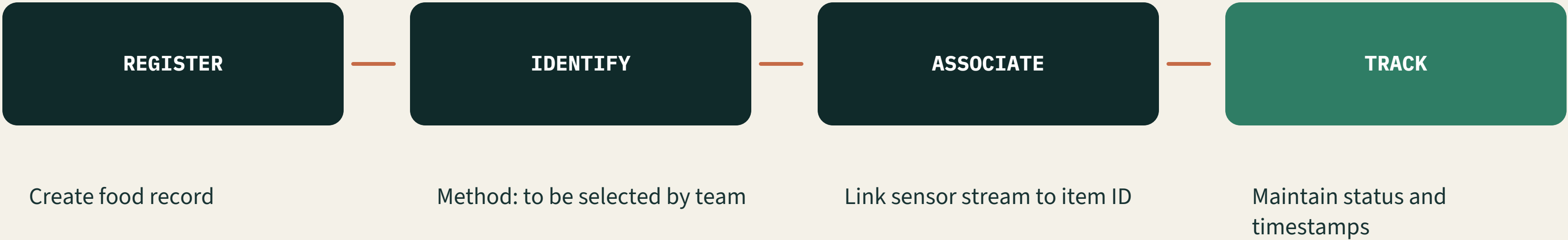
Calibration & Baseline

Prepare sensors before freshness monitoring



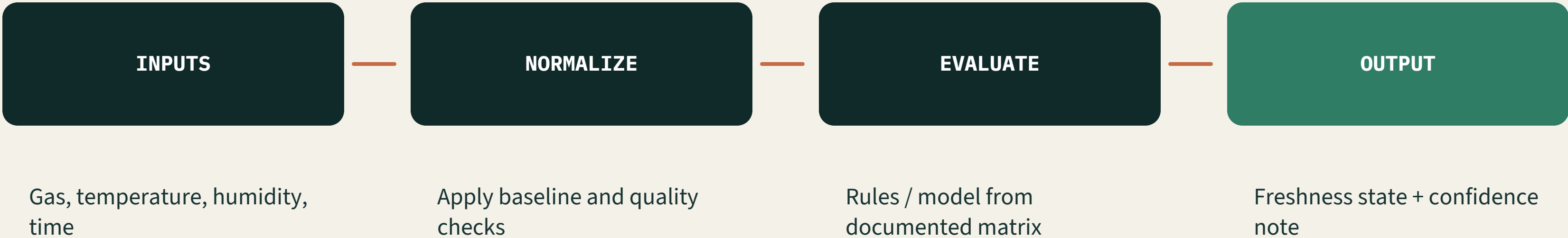
Food Registration & Identification

Create a durable identity before monitoring



Freshness Analysis Engine

Inputs, normalization and decision flow



Threshold & Decision Matrices

Document rules without inventing unspecified values

FIELD	TEAM DOCUMENTATION REQUIREMENT
Food category	Name, storage condition and expected monitoring period
Sensor input	Unit, sample interval, baseline and valid range
Threshold	Exact value: to be selected by team and justified
Decision rule	Inputs, operators, persistence window and state
Evidence	Calibration run, source and test result

Alerts & Dashboard

Consistent feedback across local and remote channels

LOCAL

Display, LED or buzzer communicates current state.

REMOTE

Send notification when connectivity and configured service permit.

DASHBOARD

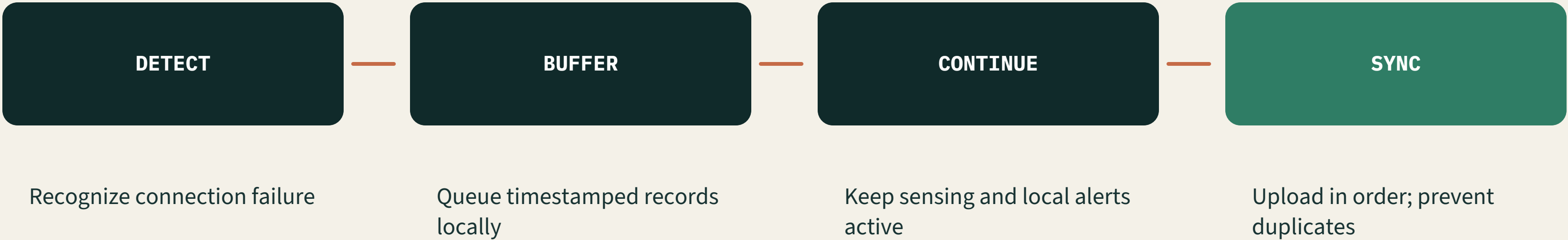
Show item, latest values, state, timestamp and connection status.

MESSAGE

Use clear severity, cause, time and recommended user check.

Offline Sync & Data Handling

Store-and-forward behavior during network loss



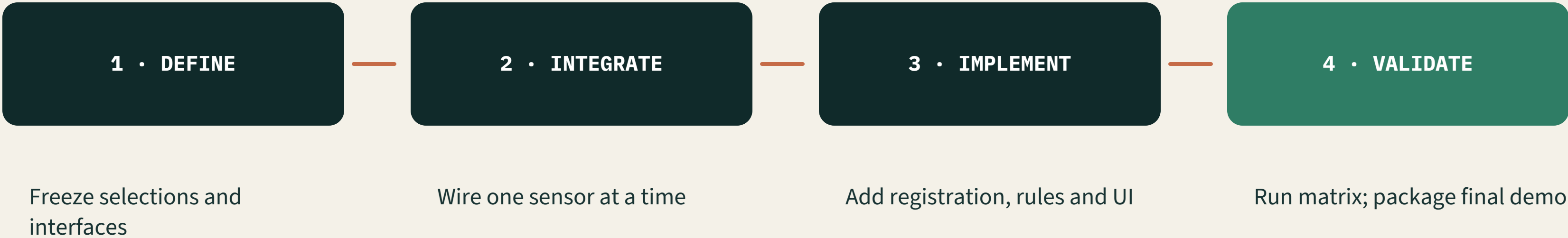
Testing & Validation

Minimum IoT test matrix

SCENARIO	EXPECTED EVIDENCE
Power-up and sensor initialization	Stable boot log; failure state visible
Known environmental change	Raw values respond in plausible direction
Food state transition	Rule and alert activate as documented
Temporary network loss	Local operation continues; records buffered
Reconnect and sync	Queued records upload once, in order

Implementation Procedure

Recommended build order



Troubleshooting

Common prototype faults and first checks

SYMPTOM	CHECK FIRST
No sensor readings	Power rails, common ground, pin map, bus address
Values fixed or implausible	Warm-up, units, range, baseline, damaged sensor
Dashboard stale	Timestamp, network status, endpoint and credentials
Repeated alerts	Threshold hysteresis / persistence: team-defined
Sync duplicates	Record ID, acknowledgement and retry logic

Submission & Demo Checklist

Evidence required for a reproducible prototype

REPORT

Architecture, decisions, calibration, matrices, results and limitations.

SOURCE ZIP

Firmware, dashboard, schema, README, dependency versions; no live secrets.

DEMO

Registration → sensing → decision → alert → offline recovery.

FINAL GATE

All placeholders resolved or explicitly marked “to be selected by team”.

PROJECT TEAM

FreshGuard

IoT-Based Smart Food
Freshness Monitor

Inferno-Net

Student 19389957

Muhammad Hamza Atif

Student 1496338

Zohaib Khan

Student 1482421

Hafiz Syed Abdullah Touheed

Syed Abdul Hadi Arshad

Student 1483763

Project identity and member details

Names · roles · IDs · supervisor: to be completed by team

Keep final documentation reproducible: record selected controller, pin mapping, thresholds, wiring, credentials handling and dependency versions.